

Bio-Capacitive Soundscapes

Organic Mixed Reality Interface

Creative Interfaces Lab, IIITD

Redefining Interaction: From Rigid to Living Interfaces



The Paradigm Shift:

Moving beyond traditional Human-Computer Interaction (HCI) towards Organic User Interfaces (OUI).



The Vision:

Transforming a static botanical environment into a responsive, multimodal instrument.



Biophilic Interaction:

replacing "dead" binary switches with living flora, utilizing the natural moisture and cellular structure of plants as conductive pathways.



The Goal:

To create a "Tactile Symphony" where the user physically connects with nature to manifest digital art.

Sound Design

The Raga & The Algorithm



A fusion of technology and tradition, utilizing **Generative Synthesis** rooted in Indian Classical Music.



Outcome: Every interaction creates a unique, non-repetitive composition that is strictly bound by the musical grammar of the Ragas.



System Architecture and Engineering



Sensing Layer:

12 unique living plants instrumented with discrete electrodes via the Bare Conductive Touch Board.



Capacitive Sensing:

The system detects precise changes in capacitance caused by the dielectric properties of the human body upon contact with the leaf surface.



Signal Processing:

Raw touch data is converted into real-time MIDI control change (CC) messages.



Routing:

Low-latency data transmission via Midify into FL Studio, ensuring the "instrument" feels responsive and instantaneous.

Implications & Conclusion



Mixed Reality

This is not just a controller; it is an **Organic Mixed Reality** experience where the physical object (the plant) remains unchanged, but its function is digitally augmented.



Applications

Potential futures in therapeutic spaces, interactive art installations, and educational tools for botany and music.



Immersive Experience

Users report a heightened sense of connectedness to the environment—the technology becomes invisible, leaving only the connection between human, nature, and sound.



"Bio-Capacitive Soundscapes proves that the future of computing doesn't have to be cold and metallic; it can be grown."